

Macroeconomic Problems

Unemployment

Unemployment

RECAP

The five (5) economic goals of Government:

1. Sustainable **economic growth**
2. **Low unemployment** (full employment)
3. **Low inflation** (price stability)
4. A more **equitable distribution of income**
5. The **efficient allocation of resources**

UnEmployment

Australian unemployment rate is stuck at 5.2 per cent, keeping 'slack' in labour market.

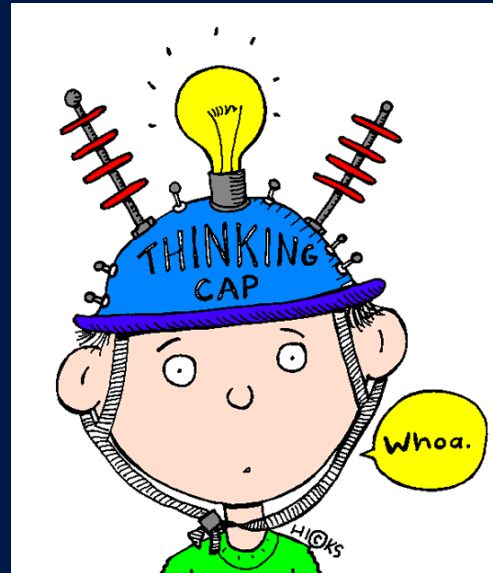
Australia's unemployment rate is stuck. And while that's good news for some people, it's terrible for anyone wanting a bigger pay cheque.

Aug 17 2019



**What is
unemployment
(UnN)?**

**How is UnN rate
calculated?**



**What are
the types of
UnN?**

**How is UnN
solved?**

**What are the
costs of UnN?**

Lesson Objectives

- Explain how unemployment is measured
- Explain why governments wish to pursue low unemployment.
- Explain the types of unemployment & its causes.
- Explain the effects of unemployment



**So what is
unemployment?**

**Who are the
unemployed ?**

The Unemployed

Who are they?

Officially, a person is unemployed if he /she is a part of the **labour force** (Economically active population) but doesn't have a job.

The Labour Force

The labour force participation rate – is the proportion of people eligible to participate in the labour force.

People who are actually participating in it, by working or looking for work.

The Labour Force

The labour force - the sum of employed and unemployed people.

The labour force is also called the working population or the economically active population.

The Economically Inactive Population

The economically inactive population comprises all persons who are neither "employed" nor "unemployed" during the short reference period used to measure "current activity".

Therefore, not working and neither looking for work

This population is split into four groups:

1. Attendant at educational instns; (full time students)
2. Retired
3. Engaged in family duties; (housewives)
4. Other economically inactive (eg. ill)

Unemployment

It is defined as the situation in which people who are willing & able to work but are unable to find work.



Unemployment

Unemployment occurs when people who are **willing** and able to work **cannot find a job**.

(Parry, G. and Kemp, S. 2018, Pg. 215)



Labour force participation

Labour force participation rate (LFPR) or
Participation Rate =

$$\frac{\text{Working population}}{\text{Total Population}} \times 100\%$$



How do we measure unemployment ?



Measuring Unemployment

How is the unemployment rate measured?

Unemployment occurs when someone is willing and able to work but does not have a paid job.

The unemployment rate is the percentage of people in the labour force who are unemployed.

Consequently, measuring the unemployment rate requires identifying who is in the labour force.

Measuring Unemployment

How is the unemployment rate measured?

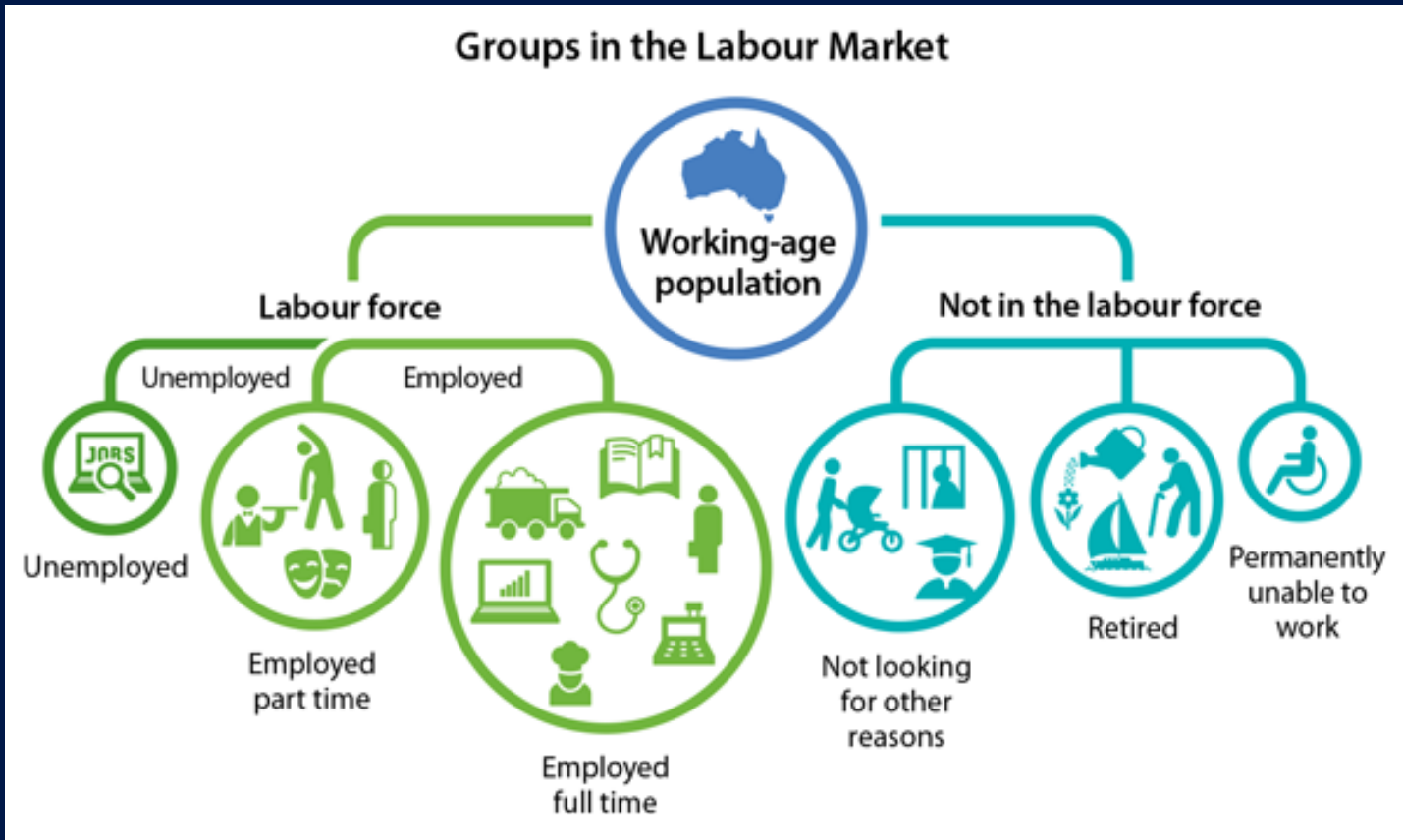
The labour force includes people who are either employed or unemployed.

Figuring out who is employed or unemployed involves making practical judgements, such as how much paid work someone needs to undertake for them to be considered as having a job, as well as actually counting how many people have jobs or not.

Measuring Unemployment

The unemployment rate is the proportion of the labour force who are willing and able to work, but have not been employed in paid work for at least one hour in the week in which they were surveyed.

Measuring Unemployment



Measuring Unemployment

Participation rate = $\frac{\text{Total Labour Force}}{\text{Population of working age}} \times 100\%$

Employment rate = $\frac{\text{No. employed}}{\text{Total Labour Force}} \times 100\%$

Unemployment rate = $\frac{\text{No. unemployed}}{\text{Total Labour Force}} \times 100\%$

Underemployment rate = $\frac{\text{No. of part time workers seeking more work}}{\text{Total Labour Force}} \times 100\%$

Measuring Unemployment

Box: Calculating the Unemployment Rate – An Example

To understand how the unemployment rate is calculated we can use an example. In this example 12.6 million people are employed and 0.7 million people are unemployed. The size of the labour force is calculated as the sum of these groups.

$$\begin{array}{rcccl} \text{Employed} & + & \text{Unemployed} & = & \text{Labour force} \\ 12.6\text{m} & + & 0.7\text{m} & = & 13.3\text{m} \end{array}$$

With the unemployment rate being the percentage of people in the labour force who are unemployed, using the numbers in our example and the equation below, the unemployment rate is calculated as 5.3 per cent.

$$\begin{array}{rcl} \text{Unemployment Rate} & = & \frac{\text{Unemployed}}{\text{Labour force}} \times 100 \\ & = & \frac{0.7\text{m}}{13.3\text{m}} \times 100 \\ & = & 5.3\% \end{array}$$

Continued over page.

Measuring Unemployment

Box: Calculating the Participation Rate – An Example

The participation rate expresses the labour force as a percentage of the working-age population. The working-age population includes Australian residents aged 15 years and over (with some exceptions, such as permanent defence force members).

Using the numbers in the example above there are 13.3 million people in the labour force and, if the working-age population is 20.0 million people, the participation rate is 66.5 per cent.

$$\begin{aligned}\text{Participation Rate} &= \frac{\text{Labour force}}{\text{Working-age population}} \times 100 \\ &= \frac{13.3\text{m}}{20.0\text{m}} \times 100 \\ &= 66.5\%\end{aligned}$$

Labour force participation typically moves with the business cycle. When businesses are hiring more workers and offering higher wages, the incentives are greater to actively look for work. In contrast, when businesses are not hiring and offering smaller wage increases, there is less of an incentive for people to look for work.

There are also structural influences affecting labour force participation that are independent of the business cycle. Some examples of trends that have affected the participation rate in the past include: more opportunities to work part time; an increase in the number of females looking for work (increased female participation); and people working for longer as they delay their retirement.

Measuring Unemployment

<https://www.ceicdata.com/en/indicator/australia/unemployment-rate>

www.abc.net.au/news/2019-06-13/unemployment-remains-steady-despite-extra-42000-jobs/11205060

Some terms to take note of....

- **Voluntary Unemployment** :- when labour chooses not to take up available jobs
- **Involuntary Unemployment** :-willing & able but cannot find job
- **Underemployment** :- employed but not being used to the maximum potential
- **Full employment** :-Situation where people who are able & willing to work can find work or where all productive resources are fully employed (not lying idle or underemployed).

Natural Rate of Unemployment

- Even at the full employment rate, there is still some form of unemployment, known as the natural rate of unemployment.
- The natural rate of unemployment is a combination of frictional, structural, and surplus unemployment.
- Even a healthy economy will have this level of unemployment because workers are always coming and going, looking for better jobs.

Natural Rate of Unemployment

- This jobless status, until they find that new job, is the natural rate of unemployment.
- It occurs at a 'normal' or 'natural' rate of between 1% and 4%.
- **The Natural Rate of Unemployment for Australia is currently 4.5%**

The Unemployment Rate

- The unemployment rate *provides an important gauge of spare capacity in the labour market and the economy more generally.*
- However, other factors also affect unemployment, which complicates its interpretation when informing monetary policy.
- Statistical methods can be used to estimate the extent to which the unemployment rate reflects spare capacity versus more enduring structural factors.

Main types of Unemployment

What are the main types of unemployment?

- There are three main types of unemployment – **cyclical**, **structural** and **frictional** unemployment.
- In practice, these cannot be measured directly, and they can often overlap, but they provide a useful way of thinking about unemployment.

Main types of Unemployment

- **Frictional** Unemployment
- **Structural** Unemployment
- **Cyclical** Unemployment
- **Seasonal** Unemployment
- **Disguised** Unemployment



Main types of Unemployment

- **Frictional** Unemployment 
- **Structural** Unemployment
- **Cyclical** Unemployment
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Frictional Unemployment

- Frictional unemployment occurs when people move between jobs in the labour market, as well as when people transition into and out of the labour force.
- Movement of workers is necessary for a flexible labour market and helps achieve an efficient allocation of labour across the economy.

Frictional Unemployment

- However, people may not find jobs immediately and need to invest time and effort in searching for the right job.
- Businesses also spend time searching for suitable candidates to fill job vacancies.
- As a result, people looking for jobs are not matched immediately with vacancies and may experience a period of temporary unemployment.

Frictional Unemployment

- This type of unemployment is generally shorter term (less than one month).
- Frictional unemployment is likely to occur at all points of the business cycle and, like structural unemployment, may not influence wages or inflation.

Frictional Unemployment

Summary

Occurs whenever there are people leaving their jobs in search of better ones voluntarily.

Associated with normal labour turnover.

Exists when the economy is at Full Employment.

Frictional Unemployment

Reasons

Imperfect market knowledge

- due to ignorance of job opportunities and rates of pay

Aggravated by geographical immobility of labour

- takes time for workers to shift to new places where there is demand for labour

Main types of Unemployment

- **Frictional** Unemployment 
- **Structural** Unemployment 
- **Cyclical** Unemployment
- **Seasonal** Unemployment
- **UnderEmployment or Disguised Unemployment**

Structural Unemployment

- Structural unemployment occurs when there is a mismatch between the jobs that are available and the people looking for work.
- This mismatch could be because jobseekers don't have the skills required to do the available jobs, or because the available jobs are a long way from the jobseekers.

Structural Unemployment

- Workers may become unemployed if they work in industries that are declining in size or have skills that could be automated as a result of large-scale technological advances.
- It may be difficult for them to find work in another industry and they may need to develop new skills or move to a region that has more opportunities.

Structural Unemployment

- For example, there has been a noticeable decline in the share of people employed in routine manual jobs over recent decades with some of these jobs being automated because of advances in technology.
- The manufacturing industry is an example of an industry that has a high share of routine manual jobs and its size in the economy has declined (both in terms of production and employment).

Structural Unemployment

- Structural unemployment tends to be longer lasting than other types of unemployment.
- This is because it can take a number of years for workers to develop new skills or move to a different region to find a job that matches their skills.
- As a result, workers who are unemployed because of structural factors are more likely to face long-term unemployment (for more than 12 months).

Structural Unemployment

- In contrast to cyclical unemployment, structural unemployment exists even when economic conditions are good.
- In theory, this type of unemployment should not directly influence wages or inflation and is best addressed through policies that focus on skills and the supply of labour.

Structural Unemployment

Summary

Occurs when there is immobility of resources (e.g. *Geographical & occupational immobility*), **change in the structure of the economy** or when there are **permanent changes** in Demand and Supply conditions.

due to:

A mismatch of skills between those possessed by the workers and those required by the industries.

Structural UnN

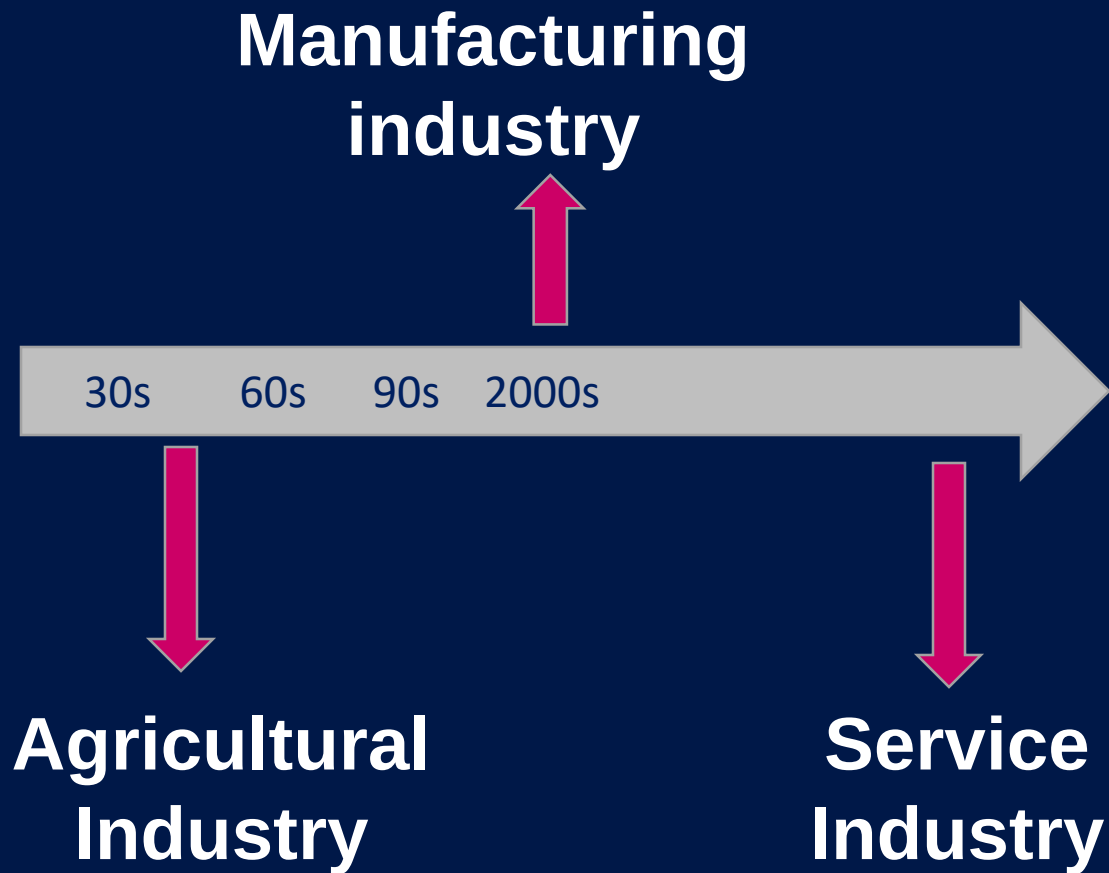
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graph TD; A[Structural UnN] --> B[Changes in DD conditions]; A --> C[Changes in structure of the economy]; A --> D[Changes in SS conditions];
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Changes in
DD conditions

Changes in
structure of the
economy

Changes in
SS
conditions

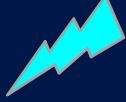
Changes in Structure of Economy



When the economy shifts to a new structure

-results in lack of skills and training

Main types of Unemployment

- **Frictional** Unemployment 
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Cyclical Unemployment

- Cyclical unemployment occurs with changes in economic activity over the business cycle.
- During an economic downturn, a shortfall of demand for goods and services results in a lack of jobs being available for those who want to work.
- Businesses experiencing weaker demand might reduce the amount of people they employ by laying off existing workers, or hiring fewer new workers.

Cyclical Unemployment

- As a result, people looking for work will also find it harder to become employed.
- The opposite situation occurs when demand strengthens.
- Cyclical unemployment is often described as being medium term in nature (one to 12 months).
- Examples can be seen in the unemployment rate rising sharply with the early 1990s recession, declining to low levels by the mid 2000s before rising again around the time of the global financial crisis.

Cyclical Unemployment

- An increase in cyclical unemployment might suggest the economy is operating below its potential.
- With more people competing for jobs, businesses might offer lower wage increases, which would contribute to lower inflation.
- Policies that stimulate aggregate demand, such as expansionary monetary policy, can help reduce this type of unemployment (because businesses experiencing stronger demand are likely to employ more people).

Cyclical Unemployment

Summary

- Relates to the *cyclical trends in growth* and production that occur within the business cycle.
- When business cycles are at their peak, cyclical unemployment will be low because total economic output is being maximized.
- When economic output falls, as measured by the gross domestic product (GDP), the business cycle is low and cyclical unemployment will rise.

Cyclical Unemployment

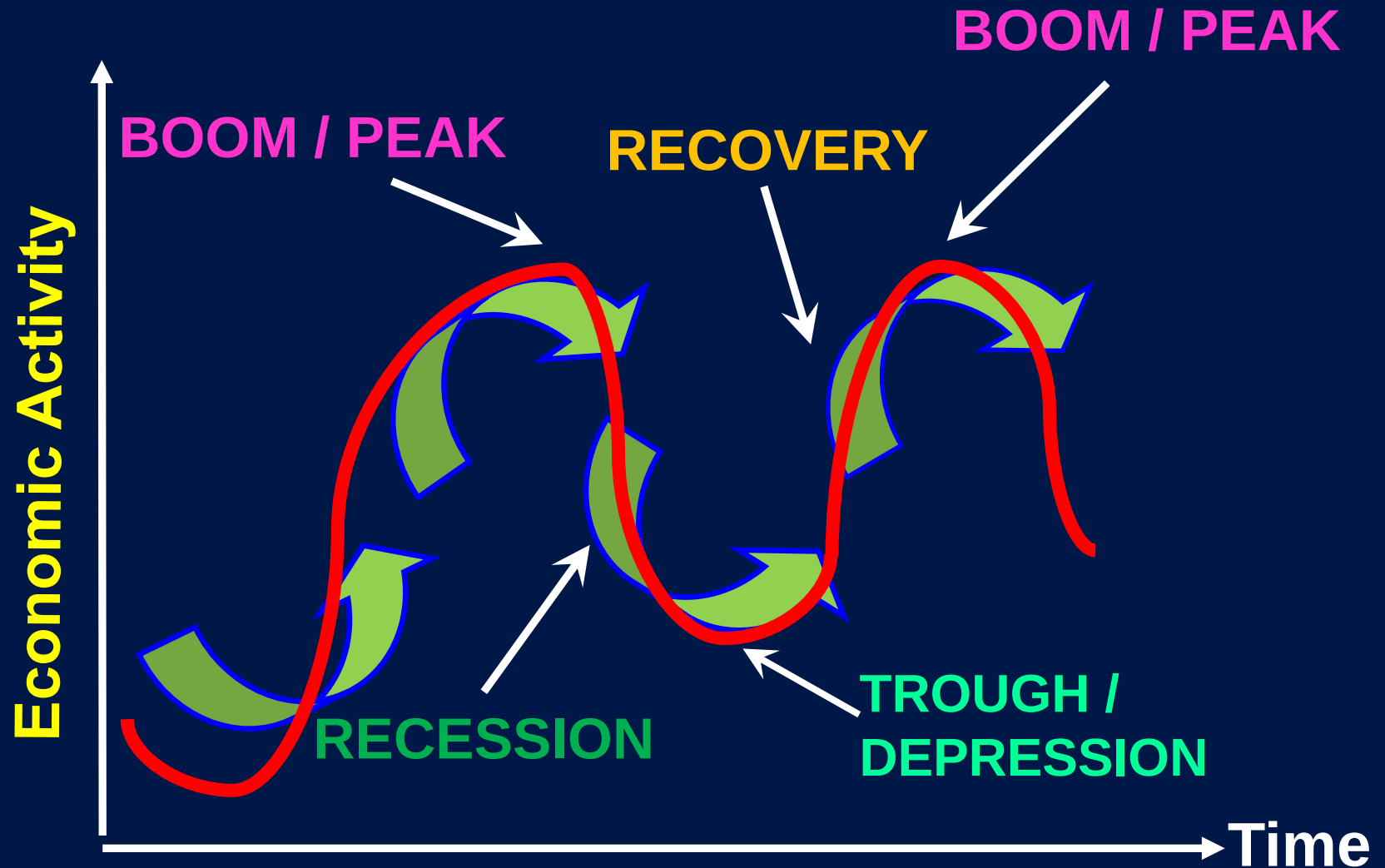
Summary

- Economists describe cyclical unemployment as the result of businesses not having enough demand for labour to employ all those who are looking for work.
- The lack of employer demand comes from a lack of spending and consumption in the overall economy.

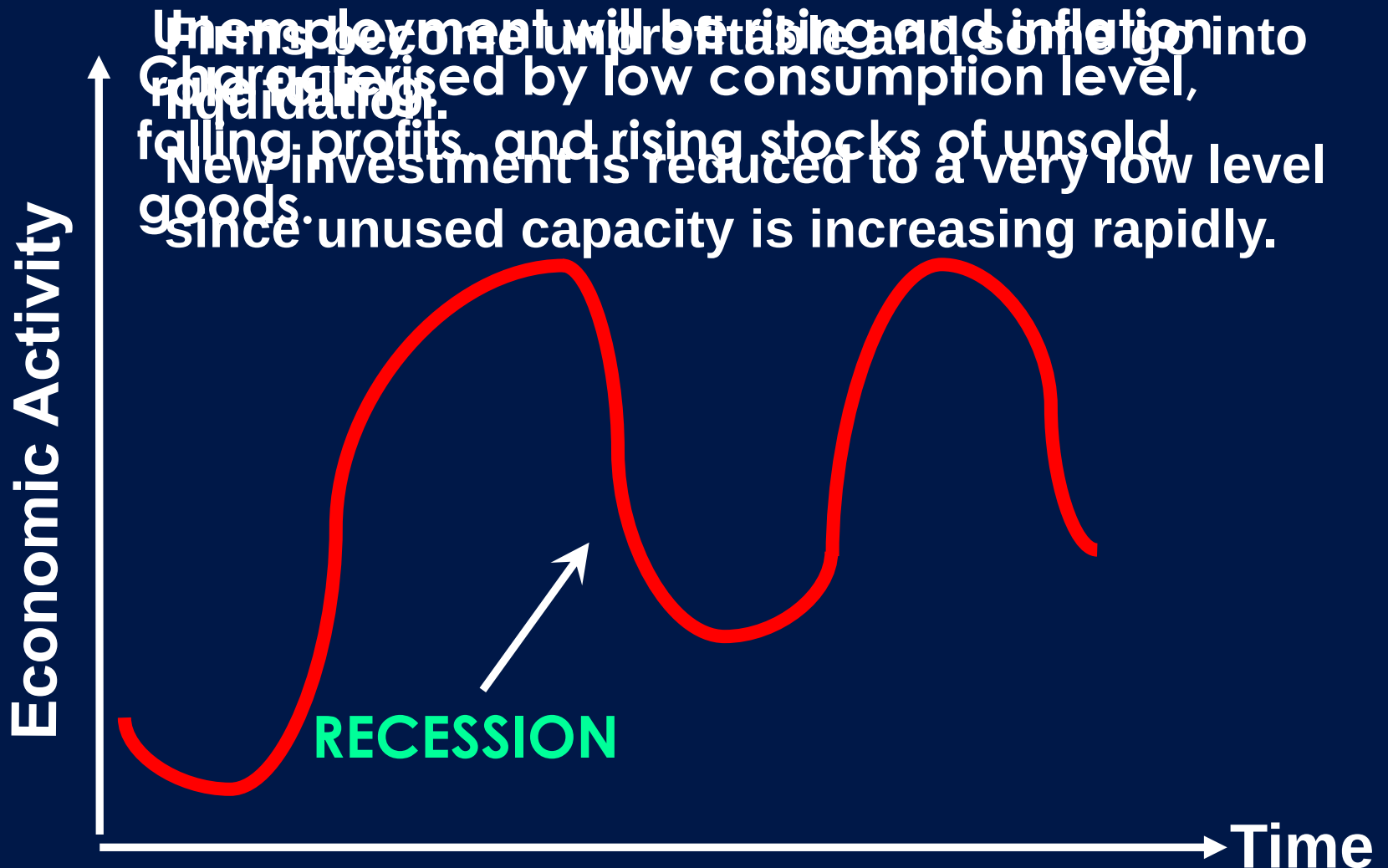
Cyclical Unemployment

- Therefore, caused by **inadequate DD** for country's gds & svs due to weak foreign and/or local demand.
- *What causes DD for gds & svs to be inadequate?*
- Due to **recession/depression** –
 - ↓ components of AD (C,I, X-M) ↓
 - aggregate demand, production and employment ↓.
 - Esp. for countries that are heavily dependent on trade.*

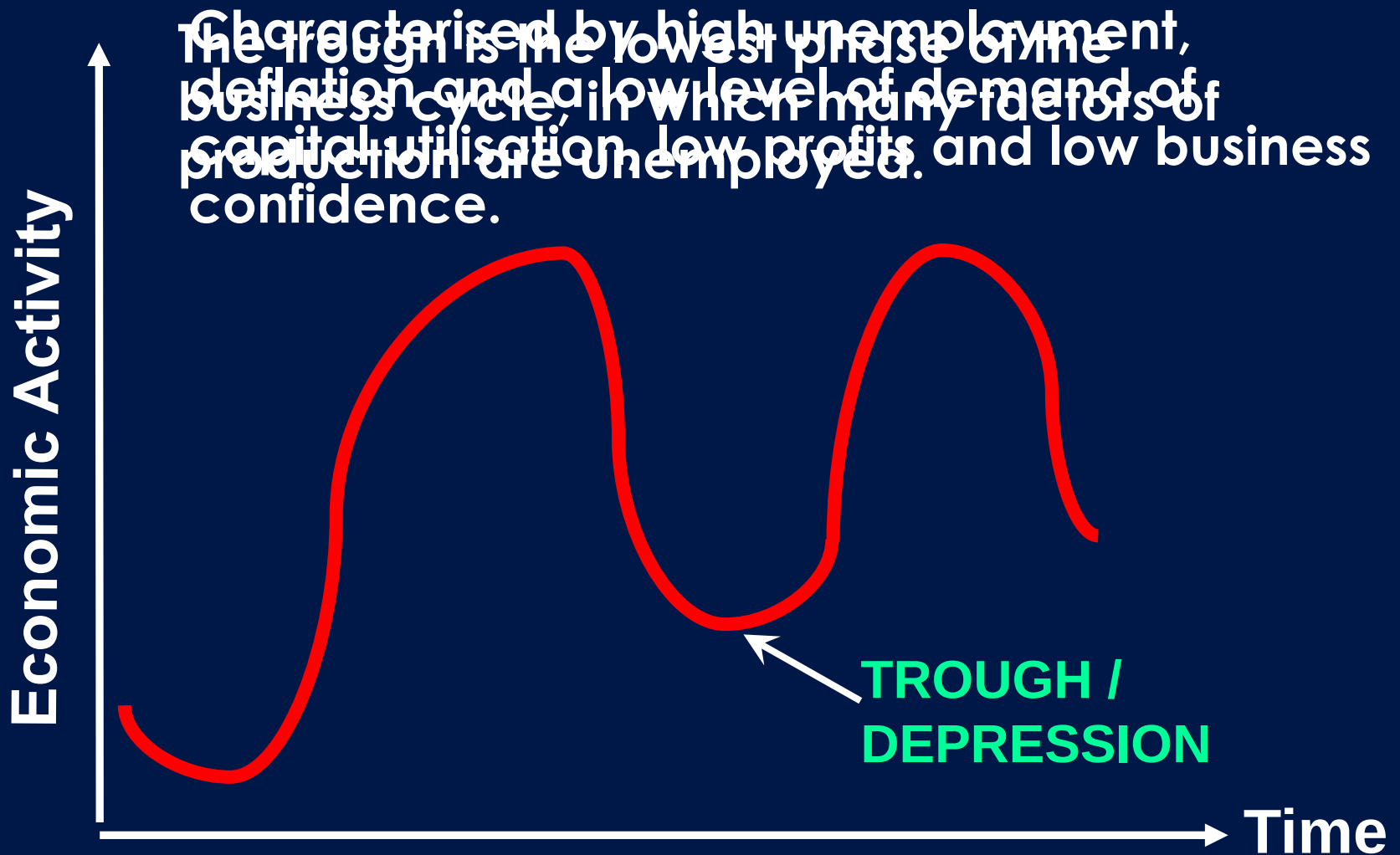
Phases of a Business Cycle



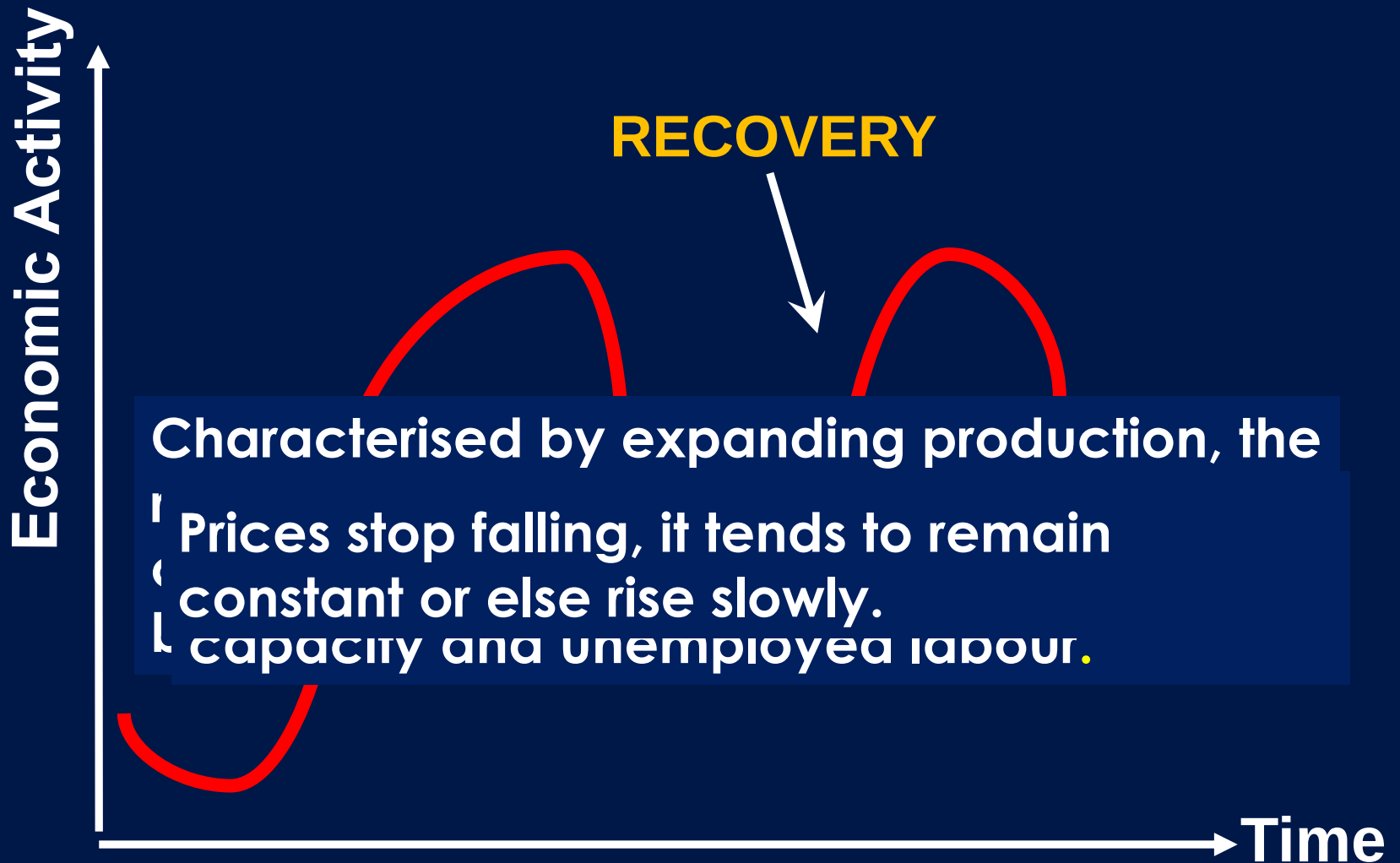
Phases of a Business Cycle



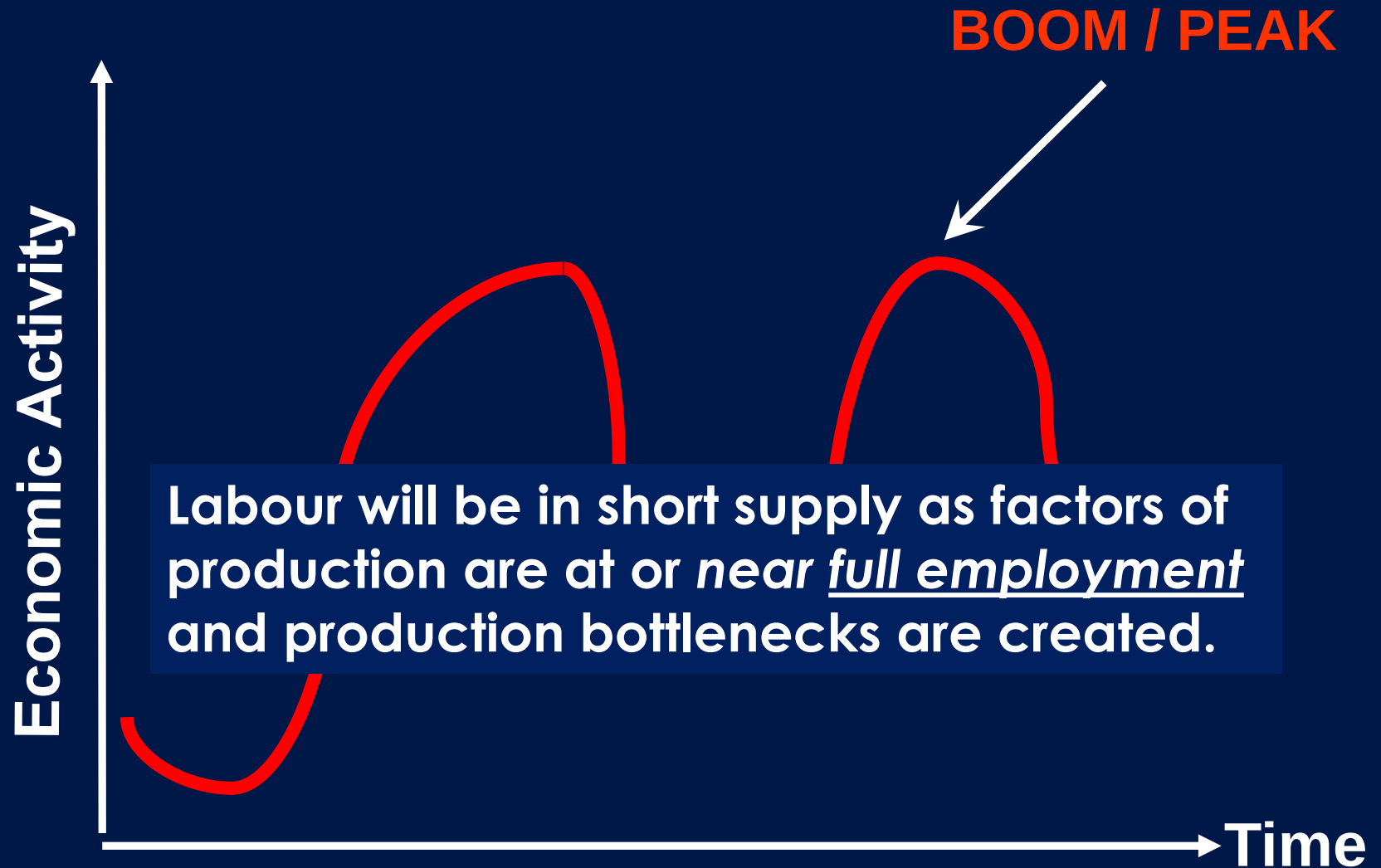
Phases of a Business Cycle



Phases of a Business Cycle

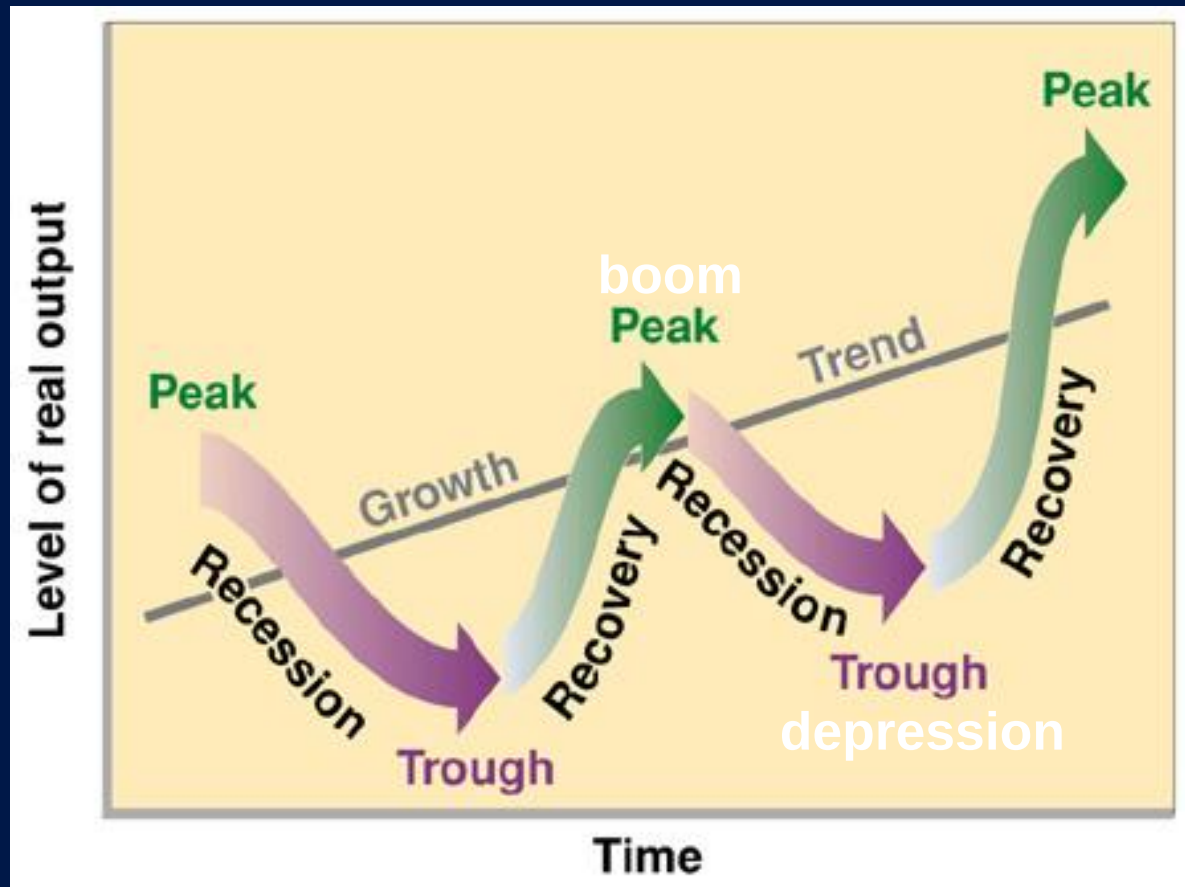


Phases of a Business Cycle



Phases of business cycle

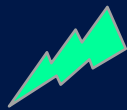
Cyclical Unemployment occurs during the **downswing** of the trade cycle, and is inherent in the cycle of business.



Unemployment: **IMPORTANT**

- These three types of unemployment are not independent of each other.
- For example, a period of high cyclical unemployment might lift structural unemployment.
- This could occur when people are unemployed for such a long period that their skills and productivity deteriorate, and they become seen as being less employable, reducing the probability that they will be hired in the future.

Main types of Unemployment

- **Frictional** Unemployment 
- **Structural** Unemployment 
- **Cyclical** Unemployment 
- **Seasonal** Unemployment 
- UnderEmployment or Disguised Unemployment

Seasonal Unemployment



Seasonal unemployment occurs when people are unemployed at particular times of the year when demand for labour is lower than usual.

Seasonal Unemployment

Problems

- those on low-income managing consumption over a year-long period can be difficult if employment is patchy. Whilst income may only come in for a few months a year, bills and rent will have to be paid every month. SoL is significantly affected if Seasonal unemployment measures are large.
- Areas of the country with high seasonal unemployment may become unattractive and local workers leave to find more stable jobs elsewhere. May impact the localized economy.

Seasonal Unemployment

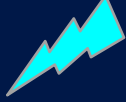
Possible Solutions

- Try to diversify the economy. This could be hard to do in touristy areas.
- Regulations which involve paying workers throughout the year, even if work is temporary.
- Government creating jobs in the off-season to improve infrastructure.



UnderEmployment

Main types of Unemployment

- **Frictional** Unemployment 
- **Structural** Unemployment 
- **Cyclical** Unemployment 
- **Seasonal** Unemployment 
- **UnderEmployment** or Disguised Unemployment 

Underemployment (Disguised) Unemployment

Underemployment or disguised unemployment occurs when workers who would like to be working full-time can only find part-time work

Underemployment (Disguised) Unemployment

This type of unemployment is due to the *inadequacy of the economy* to meet the needs of the growing labour force resulting from an *increase in population*.

Underemployment (Disguised) Unemployment

The **underemployment rate** is the proportion of people employed on a part-time basis who would like more work.

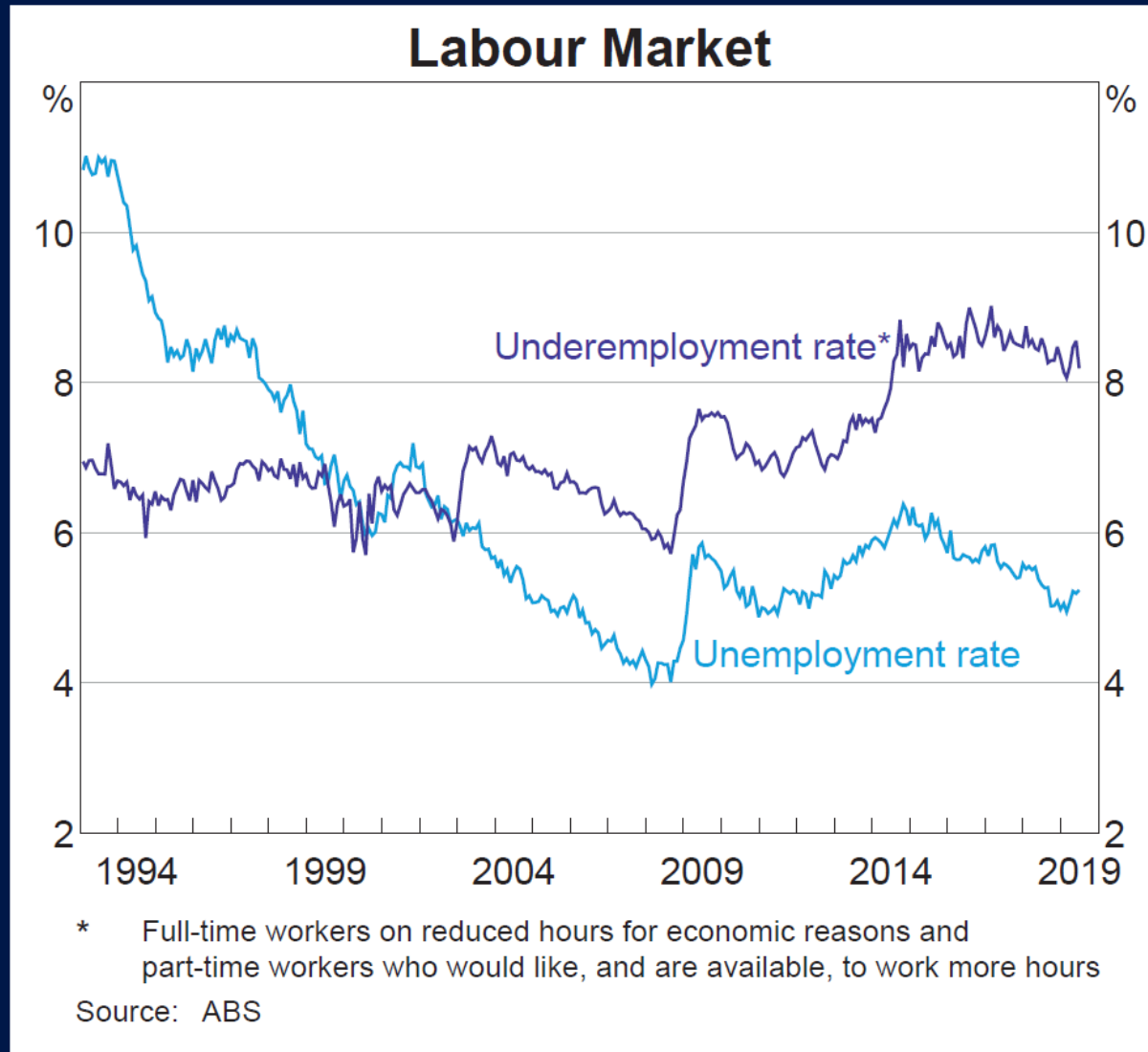
It is also an indicator of spare capacity in the economy

Underemployment (Disguised) Unemployment

Combining information about unemployment and underemployment provides a more complete picture of unused labour supply at any point in time.

Australia's unemployment rate has been trending downward in recent years while the underemployment rate has been trending upwards.

Underemployment (Disguised) Unemployment



UnderEmployment

- **Underemployment** exists frequently in developing countries whose large populations create a surplus in the labor force.
- Where more people are working than is necessary, the overall productivity of each individual drops.
- **Underemployment** is characterized by low productivity and frequently accompanies informal labor markets and agricultural labor markets, which can absorb substantial quantities of labor.

UnderEmployment - (Important: Please Read)

<https://www.abc.net.au/news/2019-05-02/underemployment-in-australia-rising-while-unemployed-rate-drops/11057794>

[https://www.aph.gov.au/About Parliament/Parliamentary Departments/Parliamentary Library/pubs/rp/rp1819/Quick Guides/Underemployment](https://www.aph.gov.au/About_Parliament/Parliamentary_Departments/Parliamentary_Library/pubs/rp/rp1819/Quick_Guides/Underemployment)

<https://theaimn.com/the-rise-of-underemployment-in-australia/>

Costs of Unemployment

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graph TD; A[Costs of Unemployment] --> B[Govt]; A --> C[Economy]; A --> D[Society]; C <--> D;
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Govt

Economy

Society

- The **Duration** of Unemployment has economic and social Costs

These Costs are most significant when long term structural unemployment is high

Costs of Unemployment

Fiscal Costs to Government

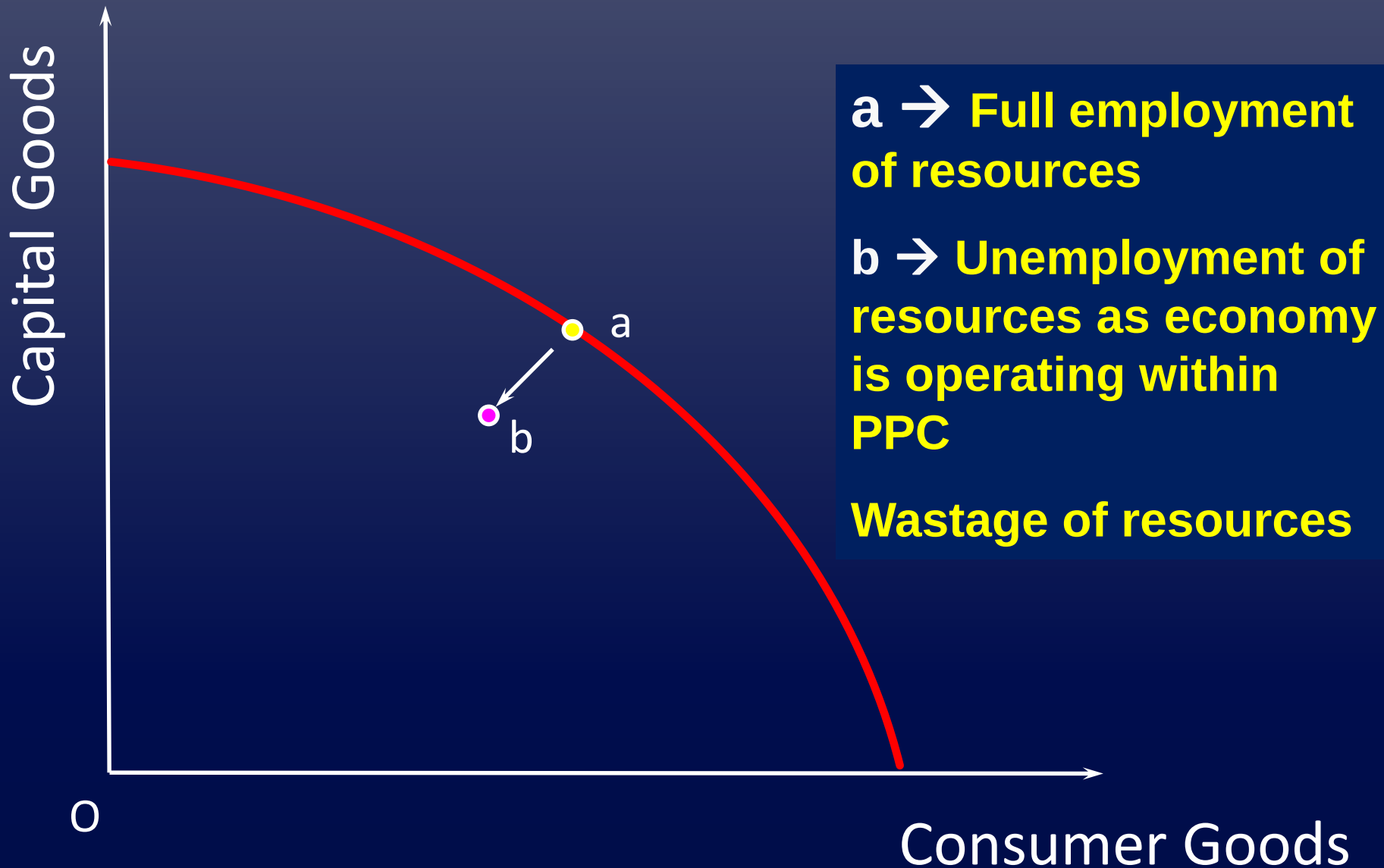
- Loss of tax revenue (both direct and indirect)
- Higher govt. expenditure incurred from unemployment benefits
- The rise in Govt Spending along with fall in tax revenues may result in higher Govt borrowing and can worsen a Govt's budget deficit

Costs of Unemployment

Economic Costs

- *'Unemployment Feeds Unemployment'*
- Wastage of Economic resources
- Reduces potential level of output (if skills become increasingly outdated in the rapidly changing job market)
- Loss in economic welfare.

Unemployment leads to wastage of resources



Costs of Unemployment

Social Costs to Society

- Social unrest (protests, crimes)
- Family breakdown
- Psychological breakdown
- The rise in Social Costs can accentuate other costs

RELATIONSHIP – UnN & INFLATION

RELATIONSHIP – UnN & Inflation

INFLATION
RATE

R2

R1

SHORT RUN
PHILIPS
CURVE

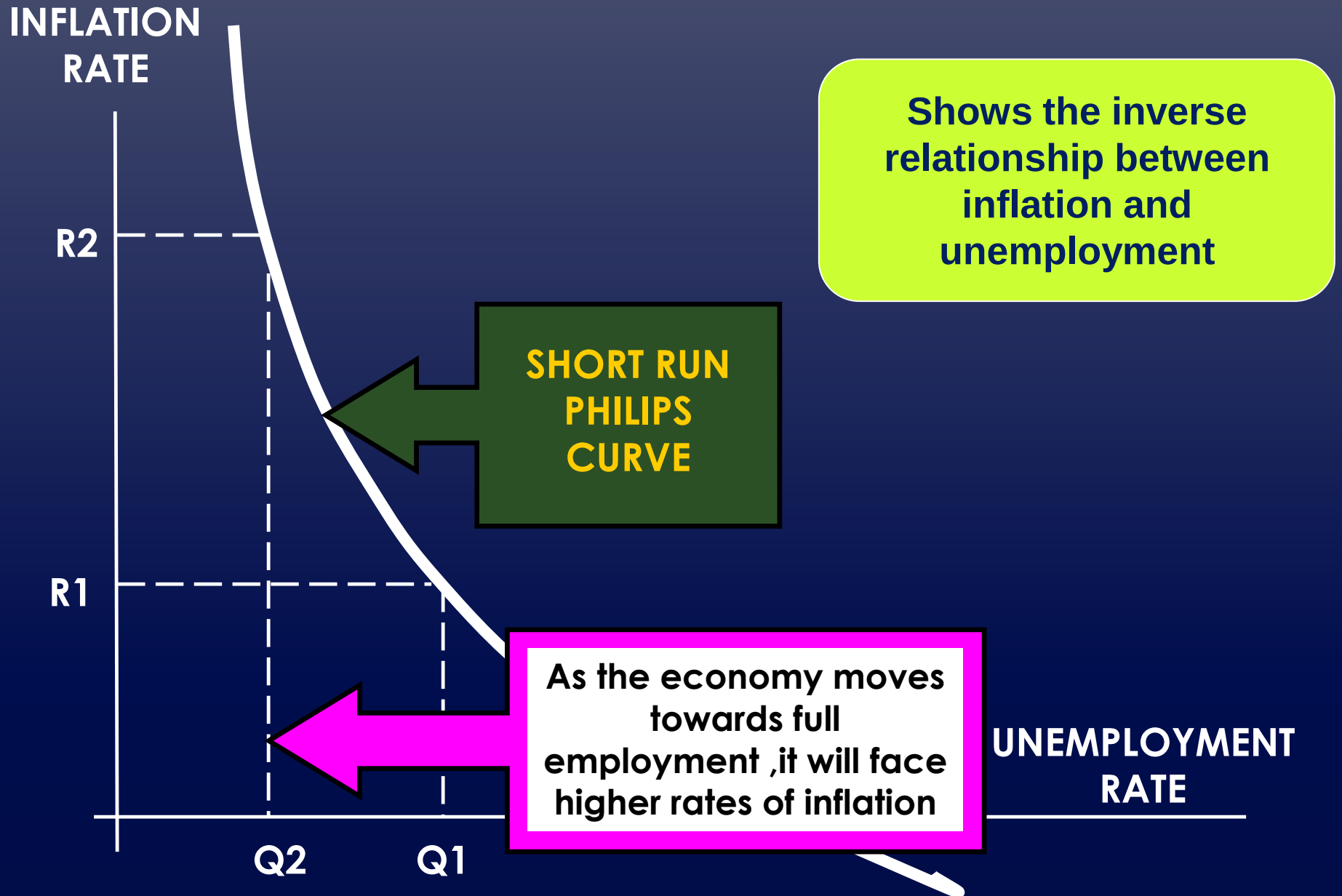
Shows the inverse
relationship between
inflation and
unemployment

As the economy moves
towards full
employment ,it will face
higher rates of inflation

UNEMPLOYMENT
RATE

Q2

Q1



RELATIONSHIP – UnN & Inflation

INFLATION
RATE

R2

R1

Q2

Q1

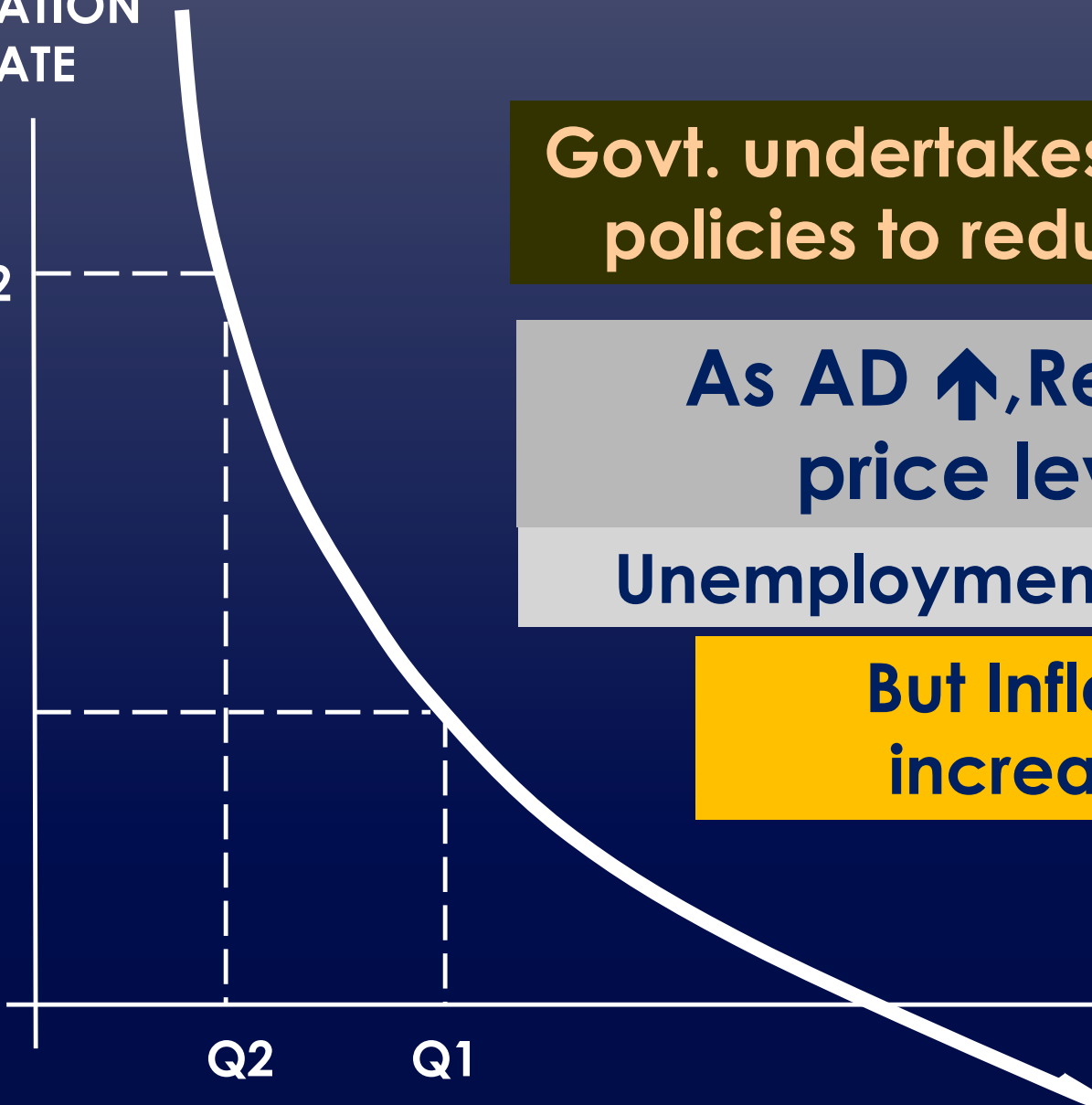
UNEMPLOYMENT
RATE

Govt. undertakes expansionary policies to reduce UnN rate.

As AD ↑, Real NY ↑,
price levels ↑

Unemployment decreases.

But Inflation
increases.



Phillips Curve in a Nutshell

This theory seeks to explain how the govt. experiences a conflict of macro objectives:

- To pursue low unN (or full employment), it is difficult to attain low rates of inflation simultaneously

STAGFLATION

Stagflation is a condition of slow economic growth and relatively high unemployment, or economic stagnation, accompanied by rising prices, or inflation. It can also be defined as inflation and a decline in gross domestic product (GDP).

STAGFLATION

When an Economy experiences the Twin evils of inflation and Slow Growth (unemployment)

The term was first used in the mid-1970s (Oil Crisis).

Stagflation is a situation in which the inflation rate is high, the economic growth rate slows down, and unemployment remains steadily high. It raises a dilemma for economic policy since actions designed to lower inflation may exacerbate unemployment, and vice versa.

STAGFLATION

CAUSES

Falling productive capacity caused by prolonged natural disaster (the LRAS shift left)

Increase in Cost of production (unfavourable supply shock e.g high price of oil).

Inappropriate macro economic policy)

END